AERIS Expert Review Packet

Anomaly

Source / metric	openaq / pm25
Observed / expected	62.2 / 8.74196
Time	2026-06-07 07:00 UTC
Location	29.8145, -95.3877
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	humidity	percent	9 / 640	59.7 to 100; mean 84.9022	100 at 2026-06-07 06:55Z; dt 5 min
asos	temperature	degC	9 / 640	23.8889 to 32.7778; mean 27.3693	26 at 2026-06-07 06:55Z; dt 5 min
asos	wind_direction	degree	9 / 640	0 to 360; mean 141.719	170 at 2026-06-07 06:55Z; dt 5 min
asos	wind_speed	m/s	9 / 640	0 to 10.8033; mean 3.6558	5.65888 at 2026-06-07 06:55Z; dt 5 min
noaa_gfs	gh_500	m	11 / 132	5853.96 to 5909.97; mean 5873.6	5867.9 at 2026-06-07 06:00Z; dt 60 min
noaa_gfs	pbl_height	m	11 / 132	44.2979 to 1876.79; mean 646.683	520.634 at 2026-06-07 06:00Z; dt 60 min
noaa_gfs	precipitable_water	mm	11 / 132	49.7819 to 59.8974; mean 55.2693	54.0192 at 2026-06-07 06:00Z; dt 60 min
noaa_gfs	surface_pressure	hPa	11 / 132	1001.91 to 1014.2; mean 1007.98	1007.01 at 2026-06-07 06:00Z; dt 60 min
noaa_gfs	t_850	degC	11 / 132	16.5548 to 20.0187; mean 18.6141	19.2674 at 2026-06-07 06:00Z; dt 60 min
noaa_gfs	u_10m	m/s	11 / 132	-4.29892 to 1.30517; mean -1.2014	-1.34029 at 2026-06-07 06:00Z; dt 60 min
noaa_gfs	v_10m	m/s	11 / 132	0.345771 to 6.44698; mean 3.1582	3.15933 at 2026-06-07 06:00Z; dt 60 min
openaq	ozone	ppm	18 / 1234	0 to 0.038; mean 0.0169	0.011 at 2026-06-07 07:00Z; dt 0 min
openaq	pm10	ug/m3	3 / 207	6 to 87; mean 33.1594	61 at 2026-06-07 07:00Z; dt 0 min
openaq	pm25	ug/m3	88 / 14157	-4.9 to 177.77; mean 6.9481	62.2 at 2026-06-07 07:00Z; dt 0 min
openweather	cloud_cover	percent	5 / 360	76 to 100; mean 98.4944	100 at 2026-06-07 06:57Z; dt 2.4 min
openweather	humidity	percent	5 / 360	67 to 96; mean 86.2778	93 at 2026-06-07 06:57Z; dt 2.4 min
openweather	precipitation	mm/h	5 / 360	0 to 7.49; mean 0.0532	0 at 2026-06-07 06:57Z; dt 2.4 min
openweather	pressure	hPa	5 / 360	1008 to 1015; mean 1010.9	1010 at 2026-06-07 06:57Z; dt 2.4 min
openweather	temperature	degC	5 / 360	24.44 to 32.69; mean 27.5534	26.44 at 2026-06-07 06:57Z; dt 2.4 min
openweather	wind_direction	degree	5 / 360	0 to 355; mean 152.492	175 at 2026-06-07 06:57Z; dt 2.4 min
openweather	wind_speed	m/s	5 / 360	0.45 to 5.94; mean 2.2147	1.79 at 2026-06-07 06:57Z; dt 2.4 min
purpleair	pm25	ug/m3	66 / 4771	0 to 286.4; mean 6.5099	8.7 at 2026-06-07 07:00Z; dt 0 min

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
sentinel5p	s5p_aer_ai_granule_available	count	7 / 7	1 to 1; mean 1	1 at 2026-06-06 20:14Z; dt 645.4 min
sentinel5p	s5p_ch4_granule_available	count	7 / 7	1 to 1; mean 1	1 at 2026-06-06 20:14Z; dt 645.4 min
sentinel5p	s5p_co_column	mol/m^2	2 / 2	0.0279382 to 0.0279482; mean 0.0279	0.0279482 at 2026-06-06 20:14Z; dt 645.4 min
sentinel5p	s5p_co_granule_available	count	7 / 7	1 to 1; mean 1	1 at 2026-06-06 20:14Z; dt 645.4 min
sentinel5p	s5p_hcho_column	mol/m^2	2 / 2	0.000118207 to 0.00012874; mean 0.0001	0.00012874 at 2026-06-06 20:14Z; dt 645.4 min
sentinel5p	s5p_hcho_granule_available	count	7 / 7	1 to 1; mean 1	1 at 2026-06-06 20:14Z; dt 645.4 min
sentinel5p	s5p_no2_column	mol/m^2	1 / 1	3.15424e-05 to 3.15424e-05; mean 0	3.15424e-05 at 2026-06-06 20:14Z; dt 645.4 min
sentinel5p	s5p_no2_granule_available	count	5 / 5	1 to 1; mean 1	1 at 2026-06-06 20:14Z; dt 645.4 min
sentinel5p	s5p_o3_granule_available	count	7 / 7	1 to 1; mean 1	1 at 2026-06-06 20:14Z; dt 645.4 min
sentinel5p	s5p_so2_column	mol/m^2	2 / 2	-9.7923e-05 to -9.174e-05; mean -0.0001	-9.7923e-05 at 2026-06-06 20:14Z; dt 645.4 min
sentinel5p	s5p_so2_granule_available	count	7 / 7	1 to 1; mean 1	1 at 2026-06-06 20:14Z; dt 645.4 min
tceq	co	ppm	3 / 215	0.1 to 0.8; mean 0.2247	0.2 at 2026-06-07 07:00Z; dt 0 min
tceq	no2	ppb	13 / 908	-2.7 to 28.3; mean 3.4452	2 at 2026-06-07 07:00Z; dt 0 min
tceq	so2	ppb	3 / 211	-0.5 to 0.7; mean -0.145	-0.2 at 2026-06-07 07:00Z; dt 0 min

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	humidity	06-05 18Z 81, 06-06 00Z 86.41, 06Z 93.76, 12Z 83.45, 18Z 82.93, 06-07 00Z 90.42, 06Z 92.7, 12Z 78.11, 18Z 73.18, 06-08 00Z 87.43, 06Z 91.28, 12Z 80.87, 18Z 65.98
asos	temperature	06-05 18Z 28.02, 06-06 00Z 26.2, 06Z 24.98, 12Z 27.9, 18Z 27.62, 06-07 00Z 27.02, 06Z 26.41, 12Z 28.25, 18Z 28.63, 06-08 00Z 27.13, 06Z 26.81, 12Z 28.79, 18Z 31.43
asos	wind_direction	06-05 18Z 122.7, 06-06 00Z 109.4, 06Z 83.14, 12Z 131.5, 18Z 154.2, 06-07 00Z 150.4, 06Z 146.7, 12Z 162.5, 18Z 164.8, 06-08 00Z 145.4, 06Z 158.3, 12Z 161.9, 18Z 165.6
asos	wind_speed	06-05 18Z 4.676, 06-06 00Z 3.164, 06Z 1.574, 12Z 2.267, 18Z 3.294, 06-07 00Z 3.468, 06Z 3.02, 12Z 5.026, 18Z 5.011, 06-08 00Z 3.639, 06Z 3.915, 12Z 4.697, 18Z 4.859
noaa_gfs	gh_500	06-06 00Z 5865, 06Z 5864, 12Z 5857, 18Z 5870, 06-07 00Z 5864, 06Z 5868, 12Z 5862, 18Z 5873, 06-08 00Z 5873, 06Z 5889, 12Z 5890, 18Z 5909
noaa_gfs	pbl_height	06-06 00Z 687.4, 06Z 239.4, 12Z 116.6, 18Z 1099, 06-07 00Z 402.7, 06Z 431.6, 12Z 477.5, 18Z 1193, 06-08 00Z 612.9, 06Z 423.5, 12Z 509.8, 18Z 1567
noaa_gfs	precipitable_water	06-06 00Z 52.65, 06Z 54.45, 12Z 55.58, 18Z 57.27, 06-07 00Z 58.09, 06Z 54.47, 12Z 56.13, 18Z 53.89, 06-08 00Z 53.94, 06Z 57.45, 12Z 55.9, 18Z 53.42
noaa_gfs	surface_pressure	06-06 00Z 1007, 06Z 1008, 12Z 1008, 18Z 1008, 06-07 00Z 1006, 06Z 1007, 12Z 1007, 18Z 1008, 06-08 00Z 1007, 06Z 1009, 12Z 1010, 18Z 1012
noaa_gfs	t_850	06-06 00Z 18.13, 06Z 17.94, 12Z 17.73, 18Z 18.3, 06-07 00Z 19.08, 06Z 19.21, 12Z 18.47, 18Z 19.08, 06-08 00Z 19.47, 06Z 19.57, 12Z 19.25, 18Z 17.13
noaa_gfs	u_10m	06-06 00Z -3.555, 06Z -1.958, 12Z -0.7138, 18Z 0.7479, 06-07 00Z -1.034, 06Z -1.048, 12Z -0.7821, 18Z -1.072, 06-08 00Z -1.02, 06Z -1.241, 12Z -1.224, 18Z -1.518

Source	Metric	Values
noaa_gfs	v_10m	06-06 00Z 2.316, 06Z 1.347, 12Z 1.509, 18Z 2.598, 06-07 00Z 3.312, 06Z 3.188, 12Z 3.295, 18Z 5.437, 06-08 00Z 3.679, 06Z 2.678, 12Z 3.57, 18Z 4.972
openaq	ozone	06-05 18Z 0.02577, 06-06 00Z 0.01893, 06Z 0.01189, 12Z 0.01376, 18Z 0.02306, 06-07 00Z 0.01591, 06Z 0.01245, 12Z 0.01811, 18Z 0.02124, 06-08 00Z 0.01469, 06Z 0.01534, 12Z 0.01342, 18Z 0.02224
openaq	pm10	06-05 18Z 12.33, 06-06 00Z 20.08, 06Z 13.44, 12Z 19.39, 18Z 13.17, 06-07 00Z 40.83, 06Z 51.06, 12Z 43.61, 18Z 41.22, 06-08 00Z 40.94, 06Z 39.5, 12Z 43.5, 18Z 45.33
openaq	pm25	06-05 18Z 3.917, 06-06 00Z 2.954, 06Z 3.064, 12Z 4.436, 18Z 4.081, 06-07 00Z 7.865, 06Z 9.41, 12Z 7.992, 18Z 6.984, 06-08 00Z 8.275, 06Z 8.576, 12Z 8.057, 18Z 6.731
openweather	cloud_cover	06-05 18Z 99.48, 06-06 00Z 98.83, 06Z 100, 12Z 99.9, 18Z 94.5, 06-07 00Z 93.43, 06Z 98.77, 12Z 97.8, 18Z 100, 06-08 00Z 99.53, 06Z 99.93, 12Z 100, 18Z 98
openweather	humidity	06-05 18Z 84.76, 06-06 00Z 87.57, 06Z 92.7, 12Z 85.87, 18Z 82.93, 06-07 00Z 90.23, 06Z 92.57, 12Z 82.47, 18Z 77, 06-08 00Z 87.13, 06Z 91.3, 12Z 82.97, 18Z 71.8
openweather	precipitation	06-05 18Z 0.0692, 06-06 00Z 0, 06Z 0, 12Z 0.1037, 18Z 0.3413, 06-07 00Z 0, 06Z 0, 12Z 0, 18Z 0.003667, 06-08 00Z 0, 06Z 0.08733, 12Z 0.045, 18Z 0
openweather	pressure	06-05 18Z 1011, 06-06 00Z 1011, 06Z 1010, 12Z 1011, 18Z 1009, 06-07 00Z 1010, 06Z 1009, 12Z 1011, 18Z 1010, 06-08 00Z 1011, 06Z 1012, 12Z 1014, 18Z 1014
openweather	temperature	06-05 18Z 28.19, 06-06 00Z 26.3, 06Z 24.96, 12Z 27.96, 18Z 28.49, 06-07 00Z 27.14, 06Z 26.34, 12Z 28.42, 18Z 28.97, 06-08 00Z 27.36, 06Z 26.87, 12Z 29.01, 18Z 32.03
openweather	wind_direction	06-05 18Z 118.3, 06-06 00Z 116.8, 06Z 136.8, 12Z 174.1, 18Z 145.4, 06-07 00Z 153.1, 06Z 159.3, 12Z 163.5, 18Z 176.2, 06-08 00Z 154, 06Z 163.4, 12Z 160.5, 18Z 168.8
openweather	wind_speed	06-05 18Z 2.514, 06-06 00Z 2.195, 06Z 1.636, 12Z 1.706, 18Z 2.441, 06-07 00Z 1.969, 06Z 1.796, 12Z 2.877, 18Z 2.716, 06-08 00Z 1.899, 06Z 2.159, 12Z 2.611, 18Z 2.86
purpleair	pm25	06-05 18Z 2.464, 06-06 00Z 2.021, 06Z 1.871, 12Z 3.853, 18Z 3.907, 06-07 00Z 9.549, 06Z 10.39, 12Z 7.687, 18Z 6.841, 06-08 00Z 9.869, 06Z 10.17, 12Z 8.726, 18Z 6.713
sentinel5p	s5p_aer_ai_granule_available	06-05 18Z 1, 06-06 18Z 1, 06-07 18Z 1
sentinel5p	s5p_ch4_granule_available	06-05 18Z 1, 06-06 18Z 1, 06-07 18Z 1
sentinel5p	s5p_co_granule_available	06-05 18Z 1, 06-06 18Z 1, 06-07 18Z 1
sentinel5p	s5p_hcho_granule_available	06-05 18Z 1, 06-06 18Z 1, 06-07 18Z 1
sentinel5p	s5p_no2_granule_available	06-05 18Z 1, 06-06 18Z 1, 06-07 18Z 1
sentinel5p	s5p_o3_granule_available	06-05 18Z 1, 06-06 18Z 1, 06-07 18Z 1
sentinel5p	s5p_so2_granule_available	06-05 18Z 1, 06-06 18Z 1, 06-07 18Z 1
tceq	co	06-05 18Z 0.2071, 06-06 00Z 0.2889, 06Z 0.2, 12Z 0.2722, 18Z 0.2778, 06-07 00Z 0.2778, 06Z 0.1333, 12Z 0.1722, 18Z 0.1933, 06-08 00Z 0.2389, 06Z 0.1667, 12Z 0.2611, 18Z 0.2167
tceq	no2	06-05 18Z 5.411, 06-06 00Z 5.392, 06Z 5.67, 12Z 3.539, 18Z 5.16, 06-07 00Z 3.372, 06Z 1.548, 12Z 1.372, 18Z 1.385, 06-08 00Z 2.668, 06Z 2.06, 12Z 3.814, 18Z 2.869
tceq	so2	06-05 18Z -0.06667, 06-06 00Z -0.05, 06Z -0.1333, 12Z -0.1056, 18Z 0.04444, 06-07 00Z -0.1714, 06Z -0.2375, 12Z -0.2278, 18Z -0.1833, 06-08 00Z -0.1722, 06Z -0.25, 12Z -0.1588, 18Z -0.26

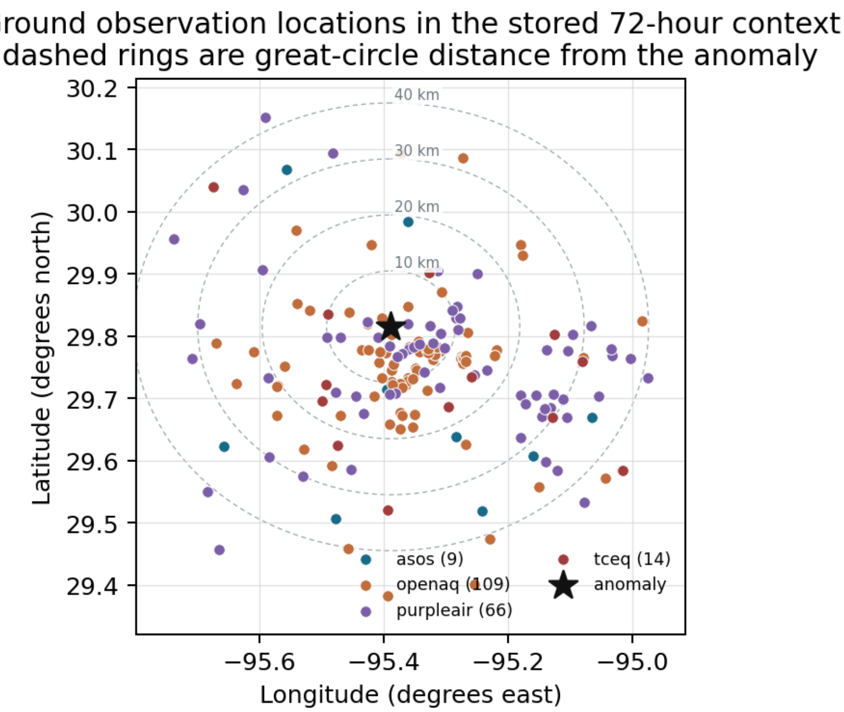
Per-site averages

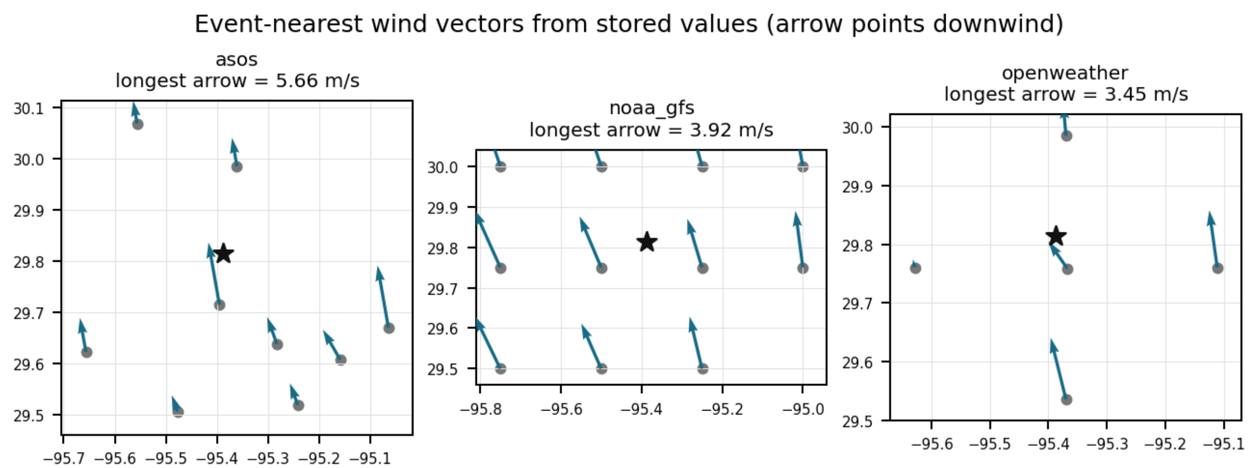
Each metric averaged at each site, with that site's distance from the event in brackets.

Source	Metric	Values
asos	humidity	MCJ (11.201 km) 88.09, IAH (19.067 km) 85.44, HOU (22.155 km) 84.18, EFD (31.938 km) 86.22, DWH (32.527 km) 84.12, SGR (33.627 km) 86.99, T41 (35.174 km) 84.24, AXH (35.363 km) 81.19, +1 more

Source	Metric	Values
asos	temperature	MCJ (11.201 km) 27.04, IAH (19.067 km) 27.21, HOU (22.155 km) 27.52, EFD (31.938 km) 27.62, DWH (32.527 km) 27.08, SGR (33.627 km) 27.29, T41 (35.174 km) 27.69, AXH (35.363 km) 27.39, +1 more
asos	wind_direction	MCJ (11.201 km) 150.7, IAH (19.067 km) 143.5, HOU (22.155 km) 160.3, EFD (31.938 km) 142.9, DWH (32.527 km) 121, SGR (33.627 km) 145.8, T41 (35.174 km) 162.1, AXH (35.363 km) 117.9, +1 more
asos	wind_speed	MCJ (11.201 km) 5.449, IAH (19.067 km) 3.601, HOU (22.155 km) 3.987, EFD (31.938 km) 4.519, DWH (32.527 km) 2.186, SGR (33.627 km) 3.815, T41 (35.174 km) 4.609, AXH (35.363 km) 1.851, +1 more
noaa_gfs	gh_500	gfs:29.75,-95.50 (12.999 km) 5873, gfs:29.75,-95.25 (15.102 km) 5874, gfs:30.00,-95.50 (23.292 km) 5872, gfs:30.00,-95.25 (24.525 km) 5873, gfs:29.75,-95.75 (35.695 km) 5872, gfs:29.50,-95.50 (36.619 km) 5874, gfs:29.50,-95.25 (37.419 km) 5875, gfs:29.75,-95.00 (38.097 km) 5876, +3 more
noaa_gfs	pbl_height	gfs:29.75,-95.50 (12.999 km) 734.4, gfs:29.75,-95.25 (15.102 km) 726.6, gfs:30.00,-95.50 (23.292 km) 774.6, gfs:30.00,-95.25 (24.525 km) 725.4, gfs:29.75,-95.75 (35.695 km) 616.5, gfs:29.50,-95.50 (36.619 km) 569.3, gfs:29.50,-95.25 (37.419 km) 608.9, gfs:29.75,-95.00 (38.097 km) 642.6, +3 more
noaa_gfs	precipitable_water	gfs:29.75,-95.50 (12.999 km) 55.37, gfs:29.75,-95.25 (15.102 km) 55.45, gfs:30.00,-95.50 (23.292 km) 55, gfs:30.00,-95.25 (24.525 km) 55.21, gfs:29.75,-95.75 (35.695 km) 55.12, gfs:29.50,-95.50 (36.619 km) 55.58, gfs:29.50,-95.25 (37.419 km) 55.48, gfs:29.75,-95.00 (38.097 km) 55.31, +3 more
noaa_gfs	surface_pressure	gfs:29.75,-95.50 (12.999 km) 1008, gfs:29.75,-95.25 (15.102 km) 1009, gfs:30.00,-95.50 (23.292 km) 1006, gfs:30.00,-95.25 (24.525 km) 1008, gfs:29.75,-95.75 (35.695 km) 1007, gfs:29.50,-95.50 (36.619 km) 1009, gfs:29.50,-95.25 (37.419 km) 1010, gfs:29.75,-95.00 (38.097 km) 1010, +3 more
noaa_gfs	t_850	gfs:29.75,-95.50 (12.999 km) 18.63, gfs:29.75,-95.25 (15.102 km) 18.53, gfs:30.00,-95.50 (23.292 km) 18.65, gfs:30.00,-95.25 (24.525 km) 18.54, gfs:29.75,-95.75 (35.695 km) 18.64, gfs:29.50,-95.50 (36.619 km) 18.64, gfs:29.50,-95.25 (37.419 km) 18.75, gfs:29.75,-95.00 (38.097 km) 18.59, +3 more
noaa_gfs	u_10m	gfs:29.75,-95.50 (12.999 km) -1.092, gfs:29.75,-95.25 (15.102 km) -1.103, gfs:30.00,-95.50 (23.292 km) -1.034, gfs:30.00,-95.25 (24.525 km) -1.075, gfs:29.75,-95.75 (35.695 km) -1.293, gfs:29.50,-95.50 (36.619 km) -1.376, gfs:29.50,-95.25 (37.419 km) -1.388, gfs:29.75,-95.00 (38.097 km) -1.201, +3 more
noaa_gfs	v_10m	gfs:29.75,-95.50 (12.999 km) 3.051, gfs:29.75,-95.25 (15.102 km) 3.053, gfs:30.00,-95.50 (23.292 km) 2.945, gfs:30.00,-95.25 (24.525 km) 2.815, gfs:29.75,-95.75 (35.695 km) 3.381, gfs:29.50,-95.50 (36.619 km) 3.131, gfs:29.50,-95.25 (37.419 km) 3.402, gfs:29.75,-95.00 (38.097 km) 3.37, +3 more
openaq	ozone	1319333 (4.693 km) 0.01897, 319 (10.034 km) 0.01549, 2046 (10.097 km) 0.01295, 310 (11.311 km) 0.0174, 3378 (15.436 km) 0.01613, 325 (16.839 km) 0.01622, 2039 (16.93 km) 0.01511, 311 (17.034 km) 0.01771, +10 more
openaq	pm10	15852419 (8.427 km) 41.4, 13380082 (15.436 km) 30.64, 271 (29.738 km) 28.61
openaq	pm25	2071327 (0.0 km) 12.35, 10107877 (1.388 km) 7.438, 13787197 (2.124 km) 7.37, 8967015 (3.659 km) 5.172, 12296914 (4.519 km) 8.864, 13787194 (4.585 km) 7.375, 8967373 (4.84 km) 7.417, 8967193 (4.94 km) 7.148, +80 more
openweather	cloud_cover	grid:center (6.48 km) 99.39, grid:north (19.034 km) 99.32, grid:west (24.006 km) 99.12, grid:east (27.362 km) 97.42, grid:south (31.042 km) 97.22
openweather	humidity	grid:center (6.48 km) 85.93, grid:north (19.034 km) 86.18, grid:west (24.006 km) 85.69, grid:east (27.362 km) 88.97, grid:south (31.042 km) 84.61
openweather	precipitation	grid:center (6.48 km) 0.04986, grid:north (19.034 km) 0.0125, grid:west (24.006 km) 0.03917, grid:east (27.362 km) 0.02944, grid:south (31.042 km) 0.1351
openweather	pressure	grid:center (6.48 km) 1011, grid:north (19.034 km) 1011, grid:west (24.006 km) 1011, grid:east (27.362 km) 1011, grid:south (31.042 km) 1011
openweather	temperature	grid:center (6.48 km) 27.68, grid:north (19.034 km) 27.43, grid:west (24.006 km) 27.56, grid:east (27.362 km) 27.41, grid:south (31.042 km) 27.69
openweather	wind_direction	grid:center (6.48 km) 149.4, grid:north (19.034 km) 132.8, grid:west (24.006 km) 153.8, grid:east (27.362 km) 160.3, grid:south (31.042 km) 166.1
openweather	wind_speed	grid:center (6.48 km) 2.123, grid:north (19.034 km) 1.964, grid:west (24.006 km) 1.46, grid:east (27.362 km) 3.584, grid:south (31.042 km) 1.944
purpleair	pm25	246011 (2.65 km) 5.299, 223813 (2.654 km) 11.2, 302037 (3.453 km) 8.34, 155367 (3.78 km) 7.766, 127451 (4.849 km) 2.175, 186303 (5.082 km) 7.384, 27821 (5.158 km) 0.3014, 296101 (5.395 km) 7.885, +58 more
tceq	co	482011052 (0.02 km) 0.3931, 482011035 (15.439 km) 0.1423, 482011039 (29.754 km) 0.1375

Source	Metric	Values
tceq	no2	482011052 (0.02 km) 7.085, 482010047 (10.026 km) 4.72, 482010024 (11.303 km) 0.6117, 482011066 (14.47 km) 4.326, 482011035 (15.439 km) 5.48, 482010416 (16.848 km) 3.825, 482010055 (17.04 km) 2.368, 482010026 (25.33 km) 6.054, +5 more
tceq	so2	482011035 (15.439 km) -0.2264, 482010051 (22.781 km) -0.2662, 482011039 (29.754 km) 0.06765





Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). The labeling guide that comes with this packet has the definitions and marking rules. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Claim 1 of 36

The contemporaneous pollutant mix favors a non-combustion or weak-combustion particle source over a major fresh combustion plume because ozone was 0.011 ppm, nearby TCEQ CO was 0.2 ppm, nearby TCEQ NO2 was 2.0 ppb, and nearby TCEQ SO2 was about -0.2 ppb at 07:00 UTC.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 2 of 36

The openaq PM2.5 measurement at 29.815, -95.388 in Houston was 62.2 ug/m3 at 2026-06-07T07:00:00+00:00, compared with an expected value of 8.741958041958041 ug/m3, corresponding to a z-score of 10.913526865502508 and a severe anomaly classification.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 3 of 36

A severe thunderstorm or tornado event occurred in the area, stirring up pollutants and reducing air quality.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 4 of 36

The air quality anomaly in Houston, Texas on June 7th, 2026 is likely caused by a combination of factors including weather patterns and local emissions.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 5 of 36

The air quality anomaly is related to PM2.5 levels.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 6 of 36

The air quality anomaly in Houston, Texas is caused by high levels of CO concentrations.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 7 of 36

An openaq sensor located at (29.815, -95.388) in Houston recorded a PM2.5 concentration spike to 62.2 ug/m3 at 2026-06-07T07:00:00+00:00, compared to an expected baseline of 8.74 ug/m3.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 8 of 36

Meteorological conditions were favorable for persistence and apparent enhancement of particulate matter because relative humidity reached 100 percent at the nearest ASOS observation near the anomaly time under cloudy nighttime conditions with modest southerly flow and a GFS planetary boundary layer height near 521 m.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 9 of 36

A coarse-plus-fine mechanically generated aerosol source such as road dust, construction activity, or other local resuspension mixed with humid aerosol is a plausible cause of the event because PM10 was simultaneously elevated while ozone, NO2, CO, and SO2 remained low, which weakens the case for a broad secondary photochemical haze or a major fresh combustion plume as the sole explanation.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 10 of 36

On June 7, 2026, at 07:00 UTC, OpenAQ recorded a severe PM2.5 concentration peak of 62.2 ug/m3 in Houston, Texas.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 11 of 36

Humid nocturnal aerosol enhancement is supported because relative humidity was extremely high near the anomaly, reaching 100 percent at the nearest asos station around 06:55 UTC and about 93 percent in openweather near 06:57 UTC, while nighttime boundary-layer heights in gfs were modest at about 431 to 521 m, conditions that favor particle accumulation and hygroscopic growth.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 12 of 36

Concurrently elevated PM10 concentrations reaching 61 ug/m3 and elevated localized carbon monoxide levels near the anomaly site support localized pollutant accumulation under light southerly winds.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 13 of 36

A low nocturnal planetary boundary layer height averaging 431 meters around 06:00 UTC restricted atmospheric vertical mixing, facilitating surface aerosol accumulation prior to the PM2.5 peak recorded by OpenAQ.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 14 of 36

NOAA GFS atmospheric model data indicated a shallow nocturnal boundary layer height of approximately 431 meters around 06:00 UTC on June 7, 2026, trapping urban emissions near the surface.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 15 of 36

The PM2.5 levels are elevated at 8.7 ug/m3, which is higher than the mean of 6.5099 ug/m3.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 16 of 36

Humid nighttime boundary-layer conditions are a plausible contributing cause of the elevated PM2.5 because relative humidity was near saturation around the event and the boundary layer was only moderately deep overnight, conditions that favor particle accumulation and hygroscopic growth or optical over-response for fine-particle sensors.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 17 of 36

PurpleAir PM2.5 sensors within the region showed an increase in 6-hour average concentrations, rising to 10.39 ug/m3 around 06:00 UTC on June 7, 2026, from pre-event 6-hour means below 4.0 ug/m3.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 18 of 36

At 2026-06-07T07:00:00+00:00, the nearest openaq ozone measurement was 0.011 ppm, the collocated tceq CO measurement was 0.2 ppm, and the nearest tceq NO2 measurement was 2.0 ppb.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 19 of 36

High ambient relative humidity exceeding 90 percent promoted secondary aerosol formation and hygroscopic growth of particulate matter during early morning hours.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 20 of 36

A localized primary particulate plume from nearby combustion or industrial activity is a plausible cause of the PM2.5 spike because the anomalous monitor reported 62.2 ug/m3 while nearby PM2.5 observations were much lower, indicating strong spatial heterogeneity consistent with a narrow source influence rather than a uniform regional episode.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 21 of 36

The air quality anomaly in Houston, Texas on June 7th, 2026 is also characterized by elevated levels of NO2.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 22 of 36

A major fresh combustion plume or regional photochemical episode is contradicted because contemporaneous ozone near the anomaly was low at 0.011 ppm in openaq, nearby tceq CO was only 0.2 ppm, nearby tceq NO2 was about 2.0 ppb, and sentinel5p showed no clear CO or SO2 enhancement in the closest available overpass, which weakens support for a large fire, industrial sulfur plume, or broad secondary pollution event.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 23 of 36

The anomaly occurred on June 7th at around 07:00:00+00:00, with the nearest sensor located approximately 2.65 km away.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 24 of 36

Regional openaq PM10 measurements elevated around the time of the anomaly, reaching a 6-hour mean of 51.06 ug/m3 at 06:00 UTC on June 7, 2026, with the nearest monitor recording 61.0 ug/m3 at 07:00 UTC.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 25 of 36

Near-saturated relative humidity levels reaching up to 100 percent at 06:55 UTC supported aerosol hygroscopic growth and mass enhancement in the absence of precipitation washout.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 26 of 36

The air quality anomaly in Houston, Texas on June 7th, 2026 is characterized by elevated levels of PM2.5.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 27 of 36

The air quality anomaly in Houston, Texas is caused by high levels of NO2 concentrations.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 28 of 36

The anomaly is a result of a combination of factors including high temperatures, low humidity, and stagnant atmospheric conditions, leading to poor air mixing and increased pollutant concentrations.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 29 of 36

A shallow nocturnal planetary boundary layer and calm surface winds trapped localized particulate emissions near the ground in Houston.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 30 of 36

ASOS meteorological observations recorded surface relative humidity levels reaching 100 percent near the anomaly location around 07:00 UTC on June 7, 2026, facilitating aerosol hygroscopic growth.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 31 of 36

A localized particulate source affecting the anomalous central Houston PM2.5 monitor is supported because the anomalous openaq PM2.5 value reached 62.2 ug/m3 at 07:00 UTC while the nearest purpleair PM2.5 reading was only 8.7 ug/m3 at the same time and the area-wide 6 h means were much lower than the anomaly, which indicates strong spatial heterogeneity inconsistent with a uniformly regional event.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 32 of 36

Anthropogenic emissions from central Houston industrial and vehicular sources accumulated under stable surface atmospheric conditions.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 33 of 36

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5 particles.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 34 of 36

The nearest openaq PM10 measurement was 61.0 ug/m3 at 2026-06-07T07:00:00+00:00, indicating elevated coarse-plus-fine particulate mass at the same time as the PM2.5 anomaly.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 35 of 36

The anomaly is caused by a nearby industrial facility releasing excessive amounts of particulate matter into the atmosphere.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 36 of 36

The openaq PM2.5 anomaly at central Houston is most consistent with a localized particulate plume rather than a regionally uniform PM2.5 event because the anomalous monitor reported 62.2 ug/m3 while the 6-hour network mean was 9.41 ug/m3 and the nearest purpleair monitor measured only 8.7 ug/m3 at the same time.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.